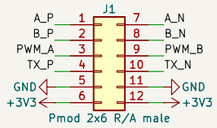
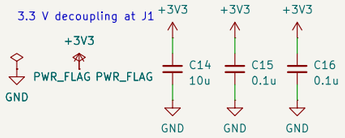
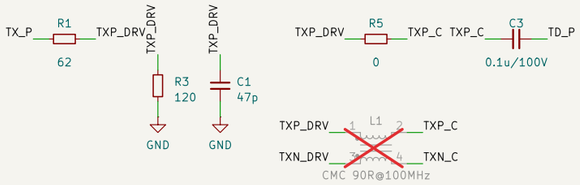
Pmod 100BASE-TX direct-attach PHY front end (transformerless) — see ../README.md

Pmod J1: plugs into Tang Primer 25K pmod0 (3.3 V bank). 1-4 top row = P, 7-10 bottom row = N.
A: 1/7 (F5/G5) B: 2/8 (G7/G8) PWM_A/B: 3/9 (H8/H7) TX_P/N: 4/10 (H5/J5) — verify by ball name.

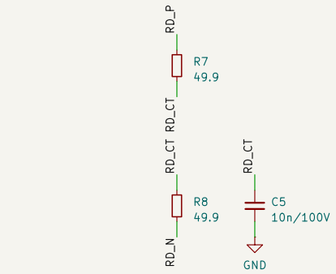




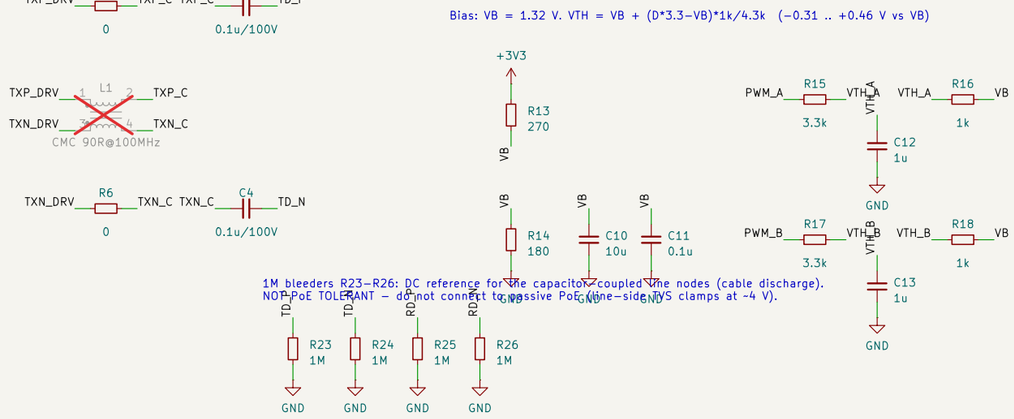
TX: LVCMOS33 x2 -> Rs 62R / Rp 120R (Thevenin 2V/50R each side) -> Cr 47pF -> [L1 CMC | R5/R6 0R]
-> 0.1uF/100V series -> RJ45 1/2. Tune Rs for 0.95-1.05 V diff, Cr for 3-5 ns rise.



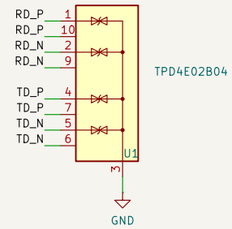
RX: RJ45 3/6 -> 2x49.9R termination, midpoint Ct to GND.
Tap caps Ca 0.1uF/100V make 4 DC-isolated comparator inputs.
A = (v > VTH_A-VB), B = (v > -(VTH_B-VB)), c = A^B = NRZI.

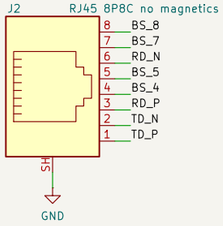


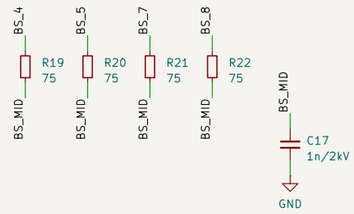
Bias: VB = 1.32 V. VTH = VB + (D*3.3-VB)*1k/4.3k (-0.31 .. +0.46 V vs VB)



U1: flow-through ESD array, NC pads 10/9/7/6 carry the same line as IO 1/2/4/5.
Line side caps rated 100 V. Bob Smith R19-R22/C17 optional.







Peer must have magnetics (normal switch / NIC). Not for board-to-board direct use.
Transformerless (capacitor coupled) 100BASE-TX front end for Tang Primer 25K

Kenta IDA

Sheet: /
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Title: Pmod 100BASE-TX direct-attach PHY

Size: A3	Date: 2026-10-10	**Rev: Rev.A**
KiCad E.D.A. 10.0.6		Id: 1/1